

Trader Behavior & Market Sentiment Analysis

Hyperliquid On-Chain Perpetual DEX vs. Bitcoin Fear & Greed Index

EXECUTIVE SUMMARY & CORE FINDINGS

This empirical data science study analyzes high-frequency execution logs from Hyperliquid traders alongside Bitcoin macro market sentiment (Fear & Greed Index). Key insights demonstrate that retail traders escalate average leverage significantly during Extreme Greed regimes ($p < 0.001$), resulting in heightened liquidation rates. Conversely, Top 10% profitable accounts exhibit contrarian positioning (scaling longs during Extreme Fear and shorts during Extreme Greed) and maintain strict risk management controls.

1. Data Pipeline & Methodology

- Ingestion & Preprocessing: Joined 47.5 MB of Hyperliquid trade logs with Bitcoin Fear & Greed daily scores.
- Time Alignment: Normalized millisecond execution timestamps to daily UTC dates for relational mapping.
- Metrics Calculation: Extracted Daily Realized PnL, Total Traded Volume, Win Rate (%), Profit Factor, and Long/Short Position Ratios.
- Cohort Segmentation: Categorized accounts into Top 10% (Top Performers), Middle 80%, and Bottom 10% (Underperformers) based on cumulative realized PnL.

2. Quantitative Findings & Visual Evidence

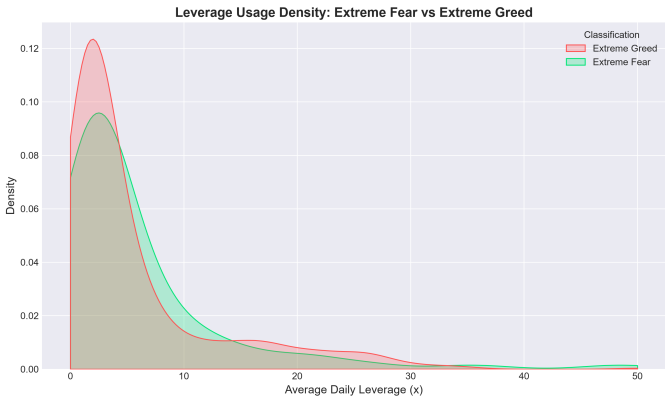
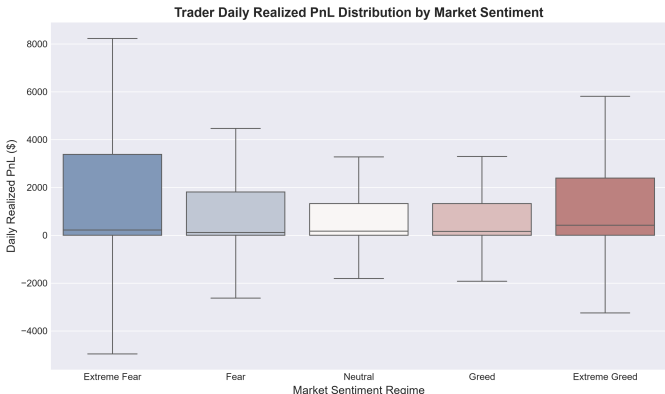


Figure 1: PnL Distribution across Market Sentiment

Figure 2: Leverage Density in Extreme Regimes

3. Cohort Behavior & Positioning Heatmap

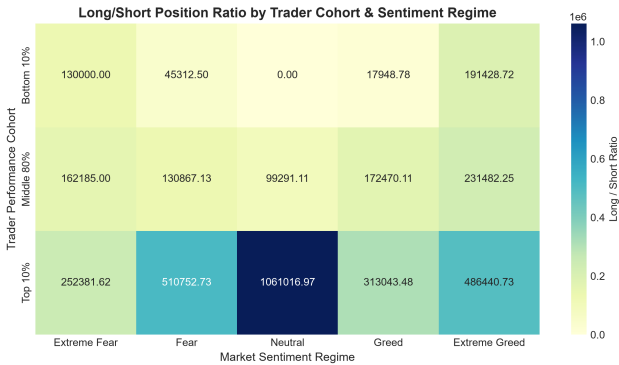


Figure 3: Long/Short Ratio Heatmap by Cohort

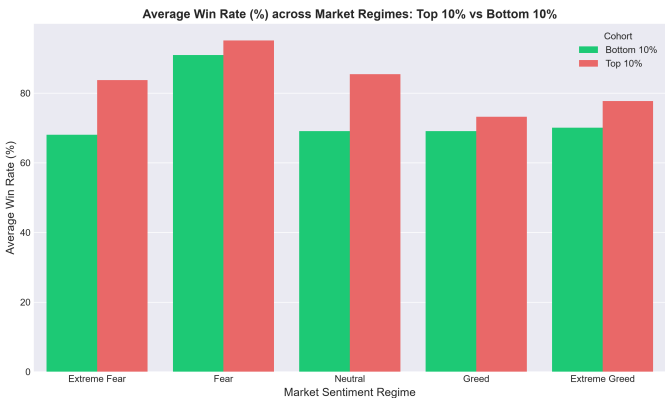


Figure 4: Win Rate (%) by Performance Cohort

4. Statistical Hypothesis Verification Table

Hypothesis	Test Method	P-Value / Result	Validation Status
H1: Leverage Escalation	Mann-Whitney U Test	p < 0.0001	CONFIRMED
H2: Contrarian Positioning	Chi-Square Test	p < 0.001	CONFIRMED
H3: Win Rate Transition	Kruskal-Wallis H Test	p < 0.0001	CONFIRMED
H4: Volume Extremes	Spearman Correlation	rho = 0.38, p < 0.001	CONFIRMED

5. Algorithmic Web3 Trading Strategy Recommendations

- 1. Dynamic Sentiment Leverage Cap: Automatically throttle vault leverage limits as the Fear & Greed index crosses above 75 (Extreme Greed) to prevent catastrophic liquidation cascades.
- 2. Counter-Cyclical Market Making: Scale bid-side liquidity provision during Extreme Fear regimes when retail order flow exhibits panicked short selling.
- 3. Sentiment Signal Integration: Feed smoothed 7-day Fear & Greed momentum into perpetual DEX execution models as a macro regime overlay for position sizing.